Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/22/2018
PI Name:	Mickens, Matthew Ph.D.		
Project Title:	Effects of Cool White LEDs on Growth and Phytonutrients of Red Romaine Lettuce when Combined with Various Monochromatic LEDs		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Organization Name:	NASA Kennedy Space Center		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899-0001	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	09/02/2015	**End Date:**	09/01/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Administration, USRA	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Massa, Gioia Ph.D. (MENTOR/ NASA Kennedy Space Center)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NASA Postdoctoral Program Fellow Growing plants in space will be an essential part of sustaining astronauts during longrange missions. To drive photosynthesis, light-emitting diodes (LEDs) have attracted attention because of their efficiency, longevity, small size, safety, and spectral variety. There have been recent interests in analyzing the effects of certain wavelengths on plant growth and phytonutrients in order to optimize crop production and quality aboard the International Space Station (ISS). With intentions to fully employ LEDs for use in space, a comparative study is proposed herein. In this work, short crop cycles of red romaine lettuce will be grown under five light treatments of cool white (CW), CW + blue (B), CW + green (G), CW + red (R), and CW + far red (FR) LEDs. This investigation will characterize and quantify biomass production, morphology, chlorophyll, and the production of key phytonutrients such as lutein, zeaxanthin, potassium, magnesium,		

	iron, and vitamin K. The crop responses to each treatment will be evaluated and used to select an LED combination that would optimize both plant growth and nutrient content.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To support food production and astronaut nutrition, my research used light-emitting diodes (LEDs) to investigate various light recipes as a strategy to improve the growth and nutrient content of leafy green crops. The scientific and technological knowledge obtained from this study can benefit space and commercial applications here on Earth.
Task Progress:	[Ed. note Oct 2018: Compiled from NPP fellow's final report submitted Sept 2018] To support food production and astronaut nutrition, my research used light-emitting diodes (LEDs) to investigate various light recipes as a strategy to improve the growth and nutrient content of leafy green crops. The primary crops studied were 'Outredgeous' red romaine lettuce and 'Rubi' red pak choi. Other crops studied but not published include Mizuna Mustard, and Chinese Cabbage. The major findings for lettuce suggest that light recipe can have profound effects on growth and morphology that could be beneficial for astronauts on long duration missions. For instance, light recipes enriched with green light were effective for rigorous growth early on the cycle for lettuce plants up 17 days old. Light recipes with high amounts of blue light resulted in small compact plants with higher nutrients such as potassium and secondary metabolites such as lutein. A recipe with light ratios similar to sunlight was observed to be most effective for increasing overall biomass accumulation. The scientific and technological knowledge obtained from this study benefits space and commercial applications here on Earth. These results have been accepted for publication in the journal, Life Sciences in Space Research. The major findings for red pak choi suggest that cabbage responds much differently than lettuce, and more favorably to narrow spectrum LED light recipes. A recipe containing red and blue only resulted in the largest plants with highest fresh mass, and highest levels of the antioxidant, anthocyanin. Unlike lettuce, green light or broad-spectrum white LEDs inhibited growth in comparison to red and blue light. Ultimately, this subsequent study demonstrated that LEDs, which offer a variety of light wavelength options, are an important capability for astronauts to have when it comes to the precise control over growing various crops. These results have been accepted for publication in the journal, Scientia Horticulturae.
Bibliography Type:	Description: (Last Updated: 02/12/2019)
Articles in Peer-reviewed Journals	Mickens MA, Skoog EJ, Reese LE, Barnwell PL, Spencer LE, Massa GD, Wheeler RM. "A strategic approach for investigating light recipes for ‘Outredgeous’ red romaine lettuce using white and monochromatic LEDs." Life Sci Space Res (Amst). 2018 Nov;19:53-62. Epub 2018 Sep 18. https://doi.org/10.1016/j.lssr.2018.09.003 ; PubMed PMID: 30482283 [Note reported originally in Oct 2018 as "In Press, Corrected Proof. Available online 18 September 2018."] , Nov-2018
Articles in Peer-reviewed Journals	Mickens MA, Torralba M, Robinson SA, Spencer LE, Romeyn MW, Massa GD, Wheeler RM. "Growth of red pak choi under red and blue, supplemented white, and artificial sunlight provided by LEDs." Sci Hortic (Amsterdam). 2019 Feb;245:200-9. Epub 2018 Oct 22. https://doi.org/10.1016/j.scienta.2018.10.023 , Feb-2019